

Toward Task-Aware Reuse of Cellular Molecular Receivers: History-Aware Prediction on Experimental FGF2–ERK Data

Ruifeng Zheng

Deutsche Telekom Chair of Communication Networks
TU Dresden, Germany

Abstract—Chemical commands delivered to a living receiver can change how it responds to later commands. We test whether the measured stimulation history of a cellular receiver improves prediction of its reporter output, using public experimental FGF2–ERK single-cell trajectories rather than artificial bit labels. A provenance-checked pilot trained on sustained and periodic stimulation reduced the 10-min-ahead root mean squared error (RMSE) on a mixed-pulse test by 8.25% relative to a current-input/response/clock baseline, but that test had already been inspected and 42.5% of its forecasts lay beyond the training time range. We therefore pre-registered an expanded analysis of 14 previously unused conditions (22 conditions and 1,565 cells in total) before downloading them. On an unseen 10-min-pulse protocol, causal-history features reduced RMSE by 8.8% relative to the clock baseline and a nested history-plus-clock model by 8.2%, yet the gain over naive persistence was only 0.7% and appeared at one of four concentrations. With five protocols available for training, history features gave the lowest RMSE in all six leave-one-protocol-out folds. We then define task-dependent receiver readiness and specify a constrained receding-horizon controller for repeated commands; the controller is not yet evaluated. The results support history-aware prediction when training stimulation is diverse, not a new biological mechanism or improved task reliability.

Index Terms—Molecular communication, cellular receivers, chemical signaling, history-aware prediction, pre-registration, FGF2, ERK.

I. INTRODUCTION

An application of molecular communication (MC) can require a chemical signal to produce a specified response in a living cell. Successful reception then means more than recovering a symbol: the receiver must execute the requested response at the right time. A previous exposure may influence the next response through transport persistence, receptor dynamics, intracellular feedback, or changes in cellular state. For a sequence of commands, an operational question is therefore when the receiver can be used again, and at what chemical and temporal cost.

We ground this question in recorded biology rather than artificial labels: FGF2 stimulation of PC-12 cells, with ERK activity measured by a FRET reporter [1]; ERK stimulation patterns are known to shape cellular outcomes [2]. The reporter is treated as a continuous receiver output, and no alphabet or cell-fate label is assigned retroactively. Section II explains why a generic learning-based controller would be a weak novelty claim.

The narrower question is whether a measured chemical-response history improves prediction beyond accessible present-state information, and whether that information can eventually reduce exposure or completion time for *repeated tasks at matched reliability*. This draft makes three bounded contributions: (i) an audited experimental input–response pilot together with a pre-registered expansion that supplies genuinely unseen protocols; (ii) a quantification of simple causal-history features that reports favorable and limiting results, including comparisons against naive persistence; and (iii) a task-dependent readiness criterion with a controller specification. Only (i) and (ii) have numerical results.

II. RELATED WORK AND POSITIONING

The FGF2 data were originally used to infer signaling architecture, including receptor interactions and feedback [1]; their nontrivial washout response is established biology, not a newly discovered memory mechanism, and frequency-dependent ERK responses are also known [2]. Postiglione *et al.* controlled mammalian gene expression and signaling with automated chemical delivery and feedback [3], and Lugagne *et al.* used a learned dynamical model for model predictive control (MPC) of thousands of single cells [4]. Nonlinear prediction, response feedback and MPC are therefore substantial precedents; the extension considered here is the feasibility and cost of a *next task conditioned on previous tasks*, not chemical feedback control itself.

Within MC, semantic systems already learn task-relevant representations [5], [6], Barros and Dey control calcium signaling and its refractory behavior [7], CELLEC adapts error control to modeled cellular noise [8], and long-term experimental MC shows that reuse and resource tradeoffs require explicit engineering [9]. These works define the comparisons a new result must satisfy rather than establishing that our formulation is unique. The candidate distinction to test is a calibrated, sequence-level exposure/time–reliability tradeoff attributable to receiver history beyond a matched-information mechanistic estimator.

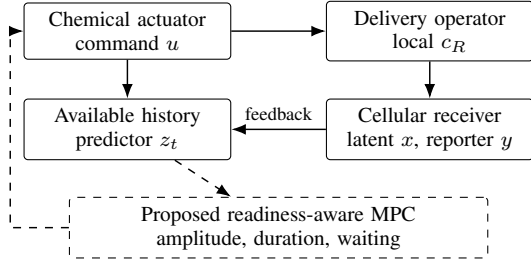


Fig. 1. MC interpretation. The completed analysis predicts the reporter from commands and available observations; local delivery, latent states and the dashed controller are not calibrated or implemented.

III. CELLULAR MC MODEL AND TASK READINESS

A. Actuation, Delivery, and Observation

Let $u(t) \geq 0$ denote the commanded FGF2 concentration in ng/ml. Delivery to a receiver is represented abstractly as

$$c_R(t) = \mathcal{T}_\vartheta[u](t), \quad \dot{x}_j = f_{\theta_j}(x_j, c_R), \quad y_j = g(x_j) + \nu_j, \quad (1)$$

where c_R is the local concentration, x_j the unobserved state of cell j , and y_j its normalized reporter. The operator $\mathcal{T}_\vartheta$ can include delivery lag and washout; neither it nor c_R is calibrated here. The FRET ratio is not a molecular count, receptor occupancy, or cell-fate measurement. For deployment, $\mathcal{H}_t$ contains executed commands and observations whose feedback has *arrived* by time t ; the offline analysis assumes each processed observation is available at its recorded time, an assumption a live experiment must replace with logged latency.

B. Task-Dependent Readiness: Proposed Definition

A prospective command m specifies an output band $\mathcal{Y}_m(t)$, a hold interval with acquisition times $\mathcal{I}_m$, a deadline, and any exclusion interval for false activation, all fixed by a separately calibrated application. At the measurement resolution, task success is

$$S_m = \mathbf{1}\left\{\bigcap_{t_i \in \mathcal{I}_m} \{y_j(t_i) \in \mathcal{Y}_m(t_i)\} \cap \mathcal{C}_m\right\}, \quad (2)$$

where $\mathcal{C}_m$ is the event that deadline and exclusion requirements hold. Individual-cell and population criteria differ and must be chosen before a policy is evaluated. For a candidate future command sequence $\mathbf{a}$, readiness is

$$R_t(m, \mathbf{a}) = \mathbb{P}(S_m = 1 \mid \mathcal{H}_t, \mathbf{a}). \quad (3)$$

Readiness is specific to a requested response and deadline: returning to a reporter baseline need not imply readiness for every next task, and successful delivery may not require complete biochemical reset. These are hypotheses to evaluate. The present data have no prespecified $\mathcal{Y}_m$ or task labels, so Eq. (2) is not applied retrospectively, and a point predictor's RMSE is not a calibrated estimate of Eq. (3).

IV. HISTORY-AWARE PREDICTION

We fit separate predictors for horizons $h = 2$ and 10 min. All learned predictors receive the same known future command: its exposure in successive 2-min bins, the concentration at $t + h$, and the horizon. Future reporter values are targets only, so this is conditional forecasting under recorded protocols, not evaluation of counterfactual commands. The input-response (IR) set contains $u_t, u_t^2, y_t, y_t^2, u_t y_t$ and the future-command features; comparators add the clock t, t^2 or the slope $(y_t - y_{t-2})/2$. The causal-history set adds this slope, the past exposure $\int_0^t u(s) ds$, and input and response filters with scales $\tau \in \{2, 10, 60\}$ min,

$$q_\tau(t) = \frac{1}{\tau} \int_0^t e^{-(t-s)/\tau} u(s) ds, \quad (4)$$

$$r_{\tau,i} = e^{-\Delta_i/\tau} r_{\tau,i-1} + (1 - e^{-\Delta_i/\tau}) y_i,$$

with $r_{\tau,0} = y_0$ and $\Delta_i = t_i - t_{i-1}$; q_τ is evaluated exactly for the piecewise-constant command. The scales are computational choices, not identified rate constants. A history-plus-clock set adds t, t^2 to the history features, giving a nested ablation of the IR-plus-clock comparator. Prediction starts after a fixed 10-min observed warmup; pre-export biological history is unknown.

Let d index training condition blocks with N_d rows each. Row weights $w_i = 1/(D_{\text{tr}} N_{d(i)})$ give every block equal weight. With features standardized by weighted training statistics, standardized row a_i and weighted target mean $\bar{y}$,

$$\hat{\beta}_\lambda = \arg \min_{\beta} \sum_{i \in \text{train}} w_i (y_{i+h} - \bar{y} - a_i^\top \beta)^2 + \lambda \|\beta\|_2^2, \quad (5)$$

with the unpenalized intercept $\bar{y}$. The penalty is selected by condition-equal validation MSE, separately per predictor and horizon; validation blocks are never added to training. The small feature set tests whether history carries usable information before a more complex architecture is justified.

V. PROPOSED RECEIVER-REUSE CONTROLLER

The proposed extension uses actions $a_k = (A_k, d_k, w_k)$: command amplitude, pulse duration, and waiting/washout time, within an experimentally calibrated set $\mathcal{A}_{\text{cal}}$ that also encodes actuator bounds, task order and deadlines. Chemical washout and cellular recovery can differ and should be measured separately. When cells share a chamber and one command, the policy must optimize a prespecified population criterion. For ordered tasks with $S_{\text{seq}} = \prod_{k=1}^K S_{m_k}$, a candidate planning problem is

$$\begin{aligned} \min_{\mathbf{a} \in \mathcal{A}_{\text{cal}}} \quad & \eta_E \mathbb{E}[E(\mathbf{a}) \mid \mathcal{H}_t] + \eta_T \mathbb{E}[T(\mathbf{a}) \mid \mathcal{H}_t] \\ \text{s.t.} \quad & \mathbb{P}(S_{\text{seq}} = 1 \mid \mathcal{H}_t, \mathbf{a}) \geq 1 - \epsilon. \end{aligned} \quad (6)$$

The constraint concerns the joint sequence event; multiplying marginal success probabilities would silently assume conditional independence across a reused receiver. The exposure $E = \int u(t) dt$ (ng·min/ml) is only a proxy: delivered mass requires calibrated flow $Q(t)$ via $\int Q(t) c_{\text{delivered}}(t) dt$, and molecule numbers further require a delivery model. A rollout

Algorithm 1: Proposed readiness-aware MPC (not executed)

Prerequisites: calibrated delivery, prespecified tasks, an action-supported rollout model and joint-event calibration.

1. Update $\mathcal{H}_t$ with commands and feedback received by t .
2. Enumerate supported amplitude/duration/wait sequences in $\mathcal{A}_{\text{cal}}$.
3. Roll out response uncertainty; estimate joint success and a calibrated lower bound $\underline{R}_t$.
4. Among sequences with $\underline{R}_t \geq 1 - \epsilon$, choose the minimum exposure/time cost; otherwise keep the reference/wait rule.
5. Execute the first action, log actuation and delivery, replan after feedback.

Fig. 2. Prospective control design; probability model, calibration, optimizer and online loop require implementation and new experiments.

model may be a mechanistic state estimator, a learned state model, or a learned residual; its distribution and task-event calibration must use separate data containing the relevant actions and outcomes.

VI. EXPERIMENTAL DATA AND EVALUATION PROTOCOL

A. Source, Audit, and Pilot Split

All numerical files come from the paper-linked `Mijan/LFNS_MSB` repository at pinned commit `5c917ab` [10]; the associated data record is cited for provenance [11]. Every file used here matches its Git blob identifier. Commands follow the authors’ model XML and pulse annotations with half-open intervals: 3/20 pulses are 3 min on every 23 min, single pulses start at 0, and mixed pulses span $[1, 4)$, $[24, 54)$ and $[114, 119)$ min. For the mixed protocol we reconstructed the 10–30 min per-cell median normalization from unnormalized exports (maximum discrepancy 1.4×10^{-4} , export rounding); other conditions retain author preprocessing. The pilot trained on the sustained and 3/20 blocks at 2.5 and 250 ng/ml, validated on single 5-min pulses, and tested on mixed pulses (606 cells, 53,998 observations). That test has since been inspected and serves only as context.

B. Pre-Registered Expansion

Before downloading any newly used response file, we froze an analysis protocol (roles, schedules, features, penalty grid $\{10^{-8}, \dots, 10^4\}$, horizons, metrics, hypotheses and data rules) and recorded its SHA-256 hash; the analysis refuses to run if the protocol changes, and it was run once. The 56 additional files all matched their upstream blob identifiers. Table I lists the 22 conditions. Training uses the sustained and 3/20 blocks at all concentrations, validation the single 5-min pulses. Test A is a new protocol (single 10-min pulse, timing from the XML). Test B is the single 60-min pulse; no XML input exists, so its $[0, 60)$ timing is inferred from the authors’ B3 simulations, which match their sustained simulations through 60 min and diverge by 64–78 min, and a ± 1 min onset sensitivity was pre-registered. Test C uses the mixed protocol at new concentrations; raw file names show it was recorded as further channels of the inspected experiment, so it shares that session. The primary hypotheses compare causal history (H1) and history plus clock (H2) with the IR-plus-clock comparator at 10 min

TABLE I

SINGLE-CELL TRAJECTORIES PER PROTOCOL AND FGF2 COMMAND AMPLITUDE (NG/ML); FRAMES PER CELL AT 2-MIN SPACING. DAGGERED MIXED BLOCKS ARE THE INSPECTED PILOT TEST; THE PILOT USED ONLY THE 2.5 AND 250 NG/ML BLOCKS.

Protocol	Role	0.25	2.5	25	250	Frames
Sustained	Train	64	59	54	71	79
3/20 pulses	Train	–	70	58	79	79
Single 5 min	Val.	105	88	92	85	69
Single 10 min	Test A	56	67	64	61	69
Single 60 min	Test B	–	66	63	38	69
Mixed pulses	Test C	87	78 [†]	84	76 [†]	130

TABLE II

PILOT CONDITION-EQUAL RMSE (NORMALIZED REPORTER UNITS). TEST AT 2 AND 10 MIN, 10-MIN VALIDATION, AND THE 10-MIN TEST SPLIT BY FORECAST ORIGIN AT THE LATEST TRAINING ORIGIN. BOLD: SMALLEST POINT ESTIMATE.

Predictor	2 min	10 min	Val.	≤ 147	> 147
Persistence	0.01588	0.03105	0.02492	0.03698	0.02047
Input only	0.04602	0.04598	0.04400	0.04725	0.04422
Input+response (IR)	0.01554	0.02960	0.02611	0.03364	0.02302
IR + clock	0.01560	0.02923	0.02648	0.03354	0.02209
IR + slope	0.01586	0.02958	0.02591	0.03386	0.02253
Causal history	0.01539	0.02682	0.02453	0.03060	0.02062

on Test A. All cells passed the rules (uniform 2-min grids, finite values). For 0.25 ng/ml the exported “unnormalized” file is identical to the normalized one, so its normalization check is uninformative; at 25 ng/ml the discrepancy is 1.4×10^{-4} .

C. Scores and Comparators

The score is the square root of the equally weighted condition MSE,

$$\text{RMSE}_{\text{cond}}(h) = \left[\frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} \frac{1}{N_{d,h}} \sum_{i \in d} (\hat{y}_{i+h} - y_{i+h})^2 \right]^{1/2}, \quad (7)$$

reported with per-condition directions. Overlapping rows and cells are not independent interventions, and experimental-day identifiers are unavailable, so no significance test or confidence interval is reported. Persistence ($\hat{y}_{t+h} = y_t$) is included because it needs no training.

VII. RESULTS

A. Pilot on the Inspected Test

Table II reproduces the pilot. At 10 min, causal history (0.02682) improves on IR plus clock (0.02923) by 8.25% and on IR by 9.39%; at 2 min the gain over IR is 0.94%. Three caveats qualify this. First, 42.5% of the 10-min test forecasts start after the latest training origin (147 min). Within that range the history gain over IR plus clock is 8.8%, but beyond it persistence (0.02047) is the best predictor, and the 2-min gain is +2.0% within and –1.8% beyond. Second, on the validation protocol history improves on persistence by only 1.6%, and IR plus clock is worse than persistence. Third, all response models select the smallest penalty (10^{-6}), so they are effectively unregularized.

TABLE III

PRE-REGISTERED 10-MIN CONDITION-EQUAL RMSE. VAL.: SINGLE 5-MIN PULSE; A: SINGLE 10-MIN PULSE (NEW PROTOCOL); B: SINGLE 60-MIN PULSE (NEW PROTOCOL, INFERRED TIMING); C: MIXED PULSES AT 0.25 AND 25 NG/ML (NEW CONCENTRATIONS, INSPECTED SESSION). BOTTOM ROWS COUNT CONDITIONS WHERE CAUSAL HISTORY HAS THE LOWER MSE.

10-min predictor	Val.	A	B	C
Persistence	0.02353	0.02287	0.02651	0.02876
Input+response (IR)	0.02535	0.02477	0.02711	0.02839
IR + clock	0.02561	0.02490	0.02709	0.02792
IR + slope	0.02503	0.02442	0.02715	0.02825
Causal history	0.02346	0.02270	0.02568	0.02778
History + clock	0.02367	0.02286	0.02624	0.03021
History beats clock	4/4	3/4	3/3	2/2
History beats persistence	1/4	1/4	2/3	1/2

B. Pre-Registered Fresh Tests

Table III reports the frozen analysis. Both primary hypotheses hold: on Test A, causal history reduces the 10-min RMSE from 0.02490 to 0.02270 (8.8%) and history plus clock to 0.02286 (8.2%), with history ahead of IR plus clock at three of four concentrations. The comparison with persistence is less favorable. The current-state models, trained on two protocols, are all worse than persistence on Tests A and B, and causal history improves on persistence by only 0.7% on Test A (one of four concentrations) and 3.1% on Test B (two of three). On Test B the ordering is unchanged under the ± 1 min onset shifts (history 0.02530 and 0.02614 against persistence 0.02651). On Test C, history and IR plus clock are nearly equal (0.02778 versus 0.02792), and history plus clock degrades to 0.03021: before the latest training origin the current-state models are slightly better (IR 0.03216 versus history 0.03284), whereas beyond it causal history is best and the extrapolated clock terms hurt. At 2 min on Test A, causal history improves on IR plus clock by 1.9% (four of four concentrations). Penalty selection again reaches the grid edge (10^{-8} for every 10-min response model).

C. Leave-One-Protocol-Out Robustness

The pre-registered robustness analysis retrains on five protocol groups and tests on the sixth, with penalties fixed at the primary selection (Table IV). History features, with or without clock terms, give the lowest RMSE in all six folds, improving on persistence by 5.8–13.6%. In particular, the 3/20 fold improves from 0.04076 (persistence) and 0.04306 (IR plus clock) to 0.03521. Compared with Table III, this indicates that the history advantage over persistence depends on how diverse the training stimulation is; it remains a protocol-level, single-session-per-protocol result.

D. Command-Only Models and Mechanistic Reference

For population means without reporter feedback, a positive single-lag model (validation time constant 13.16 min) and a ridge filter bank with scales 1–128 min were fit on the pilot split. On 0–180 min (Fig. 3, Table V), the heavily regularized bank improves on a constant training mean by only 0.9%

TABLE IV
LEAVE-ONE-PROTOCOL-OUT 10-MIN RMSE; INSPECTED MIXED BLOCKS EXCLUDED FROM TRAINING.

Held-out protocol	Persist.	+ clock	History	Hist.+clock
Sustained	0.02183	0.02135	0.02001	0.02000
3/20 pulses	0.04076	0.04306	0.03521	0.03583
Single 5 min	0.02353	0.02315	0.02182	0.02187
Single 10 min	0.02287	0.02247	0.02130	0.02133
Single 60 min	0.02651	0.02750	0.02497	0.02498
Mixed, new conc.	0.02876	0.02743	0.02699	0.02734

TABLE V
POPULATION-MEAN OPEN-LOOP RMSE ON 0–180 MIN (PILOT SPLIT).

Population-mean model	2.5 ng/ml	250 ng/ml
Positive single lag	0.05395	0.05675
Stable filter bank	0.02071	0.02245
Training-mean constant	0.02091	0.02306
Author B3 reference	0.01146	0.01362

and 2.7%, and the single lag is worse than the constant; the mean of the original authors' 1,000 B3 predictive samples is much closer. B3 used mixed and single-pulse conditions in its architecture selection, so it is a selection-exposed reference, not a blind benchmark, and population-mean errors are not comparable with the individual forecasts above. After the long pulse the mean reporter still rises from 54 to 66 min with zero command (from 1.0773 to 1.0958 at 2.5 ng/ml and from 1.0165 to 1.0742 at 250 ng/ml), which a nonnegative single lag cannot reproduce; the source biology already reports such washout behavior [1].

E. Exploratory Matched History

Pairing, within mixed-protocol cells, early washout after the short pulse (6–12 min) with later washout after the long pulse (56–102 min) at similar reporter values (tolerance 0.005), the later-minus-earlier 10-min target is positive at 2.5 ng/ml (60/78 cells, +0.0144; slope-matched 14/78, +0.0202) but negative at 250 ng/ml, shrinking to -0.0012 when slope-matched (17/76). The windows were chosen after viewing the data and post-stimulation matching does not identify a causal effect, so this is no evidence of a new memory mechanism.

VIII. DISCUSSION AND REQUIRED VALIDATION

The question is molecular because the task involves non-negative chemical actuation, delivery and washout dynamics, a nonlinear receiver, and history-dependent availability: waiting is a control action with a time cost, and exposure has its own cost. The evidence so far is predictive. Fresh protocols confirm that history features beat a current-state-plus-clock model, but against persistence the advantage appears only when training covers diverse stimulation, and the long mixed protocol exposes clock extrapolation. None of this separates delivery memory from intracellular memory or establishes task reliability.

A decisive experiment should randomize short versus long preconditioning at the independent chamber/session level,

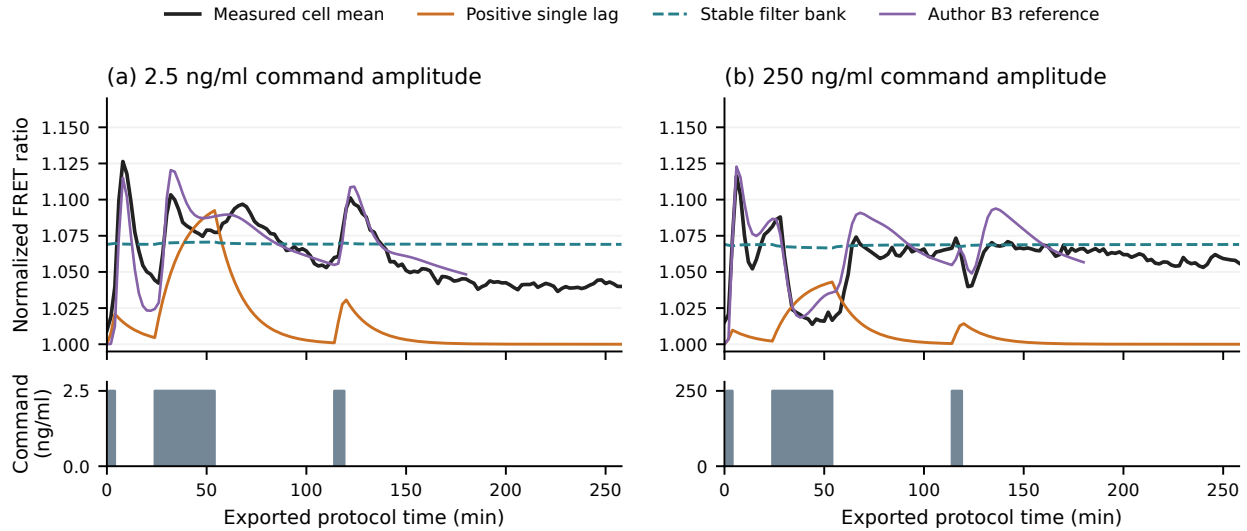


Fig. 3. Mixed-protocol cell means at the inspected concentrations with command-only models and the original authors' B3 predictive mean (ends at 180 min; selection-exposed reference), commanded pulses below. No local-concentration trace is available.

counterbalance protocol time and cell age, and then apply a common second command after candidate waits (6, 12, 24 and 48 min with a 5-min probe are pilot candidates). Local concentration, flow, actuation timestamps, feedback arrival and all failures must be logged, and task bands fixed on separate calibration data. The central comparison concerns at least two ordered commands: fixed waiting, reporter-threshold reset, a B3-based state estimator with MPC, current-state learning and history learning, all with the same observations and budget, scored by exposure and completion time at matched sequence reliability. A targeted audit of 18 neighboring works, with recorded full-text gaps, accompanies the package; it is no proof of absence and supports no priority claim. If history adds no independent-day benefit, or a mechanistic estimator reaches the same reliability and cost, the claim should be narrowed.

IX. CONCLUSION

On experimental FGF2-ERK trajectories, simple causal-history features improved 10-min prediction over a current-state-plus-clock model in a pilot and in a pre-registered test on unseen protocols, but their advantage over naive persistence was small unless training covered diverse stimulation. We formulated task-dependent readiness and a receding-horizon controller for repeated use of a cellular receiver; demonstrating its value requires calibrated delivery, prespecified tasks, supported interventions and new independent-session experiments.

DATA AND CODE AVAILABILITY

The package contains the manuscript, frozen results, the pre-registered protocol with its hash record, source manifests and verification scripts; one command re-runs the analysis and checks the reported values against a frozen reference. Sources are credited in [1], [10], [11]; author files keep their attribution and receive no new license.

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